

A member of MMC Group

Johor Port Berhad

Software Test Description (STD)

Unsafe Conditions and Unsafe Acts System (UCUX)

Application unit
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1. Introduction

The testing phase is a critical step following the development of the Unsafe Conditions Unsafe Acts (UCUX) system, aimed at rigorously verifying system functionality and stability. The fundamental objective of system testing is to test and identify errors or problems that may occur during the use of the system. This process is vital to identify code issues before the system is fully deployed and to ensure the developed application aligns with the original goals and objectives of the project, ultimately enhancing the overall effectiveness of UCUX.

2. Purpose

The main objectives of these documents are:

- To provide detailed test cases for black box testing that include expected and actual outputs.
- To provide test modules for test scenarios to obtain user perspectives or feedback on the test scenarios.

3. Scope

The Johor Port Unsafe Conditions Unsafe Acts (UCUX) System will undergo two types of testing, namely black box testing and usability testing. Black box testing aims to test the functionality of the system, while usability testing aims to assess whether the system is easy to use or not based on the user's perspective.

4. Black-Box Testing

This section explains the Black-Box testing carried out on the UCUX system. Black-Box testing is carried out to test the functions of each module in the system to ensure whether the actual results of the function are in line with the expected results. This test is carried out after the development of the system is completed and aims to identify any errors in the functions provided.

4.1 Login Test Case

This test case authenticates an existing user. The primary inputs are a username and a password.

Test Case ID	Test Description	Test Data	Expected Result
TC-LOG-01	Positive Path: Login with valid, existing credentials.	User: testuser, Pass: Password123!	User is successfully logged in and redirected to the dashboard/homepage.
TC-LOG-02	Negative Path: Login with a valid username but an incorrect password.	User: testuser, Pass: WrongPassword	Error message: "Invalid username or password." User remains on the login page.
TC-LOG-03	Negative Path: Login with a username that does not exist.	User: nonexistent, Pass: anyPassword	Error message: "Invalid username or password." User remains on the login page.
TC-LOG-04	Negative Path: Login with a valid username but a blank password.	User: testuser, Pass: (blank)	Error message: "Password is required."
TC-LOG-05	Negative Path: Login with a blank username and a valid password.	User: (blank), Pass: Password123!	Error message: "Username is required."
TC-LOG-06	Negative Path: Login with both username and password fields blank.	User: (blank), Pass: (blank)	Error messages for both missing fields.
TC-LOG-07	Case Sensitivity Test: Login with the correct username, but with the wrong case.	User: TestUser, Pass: Password123!	Error message: "Invalid username or password." (Assumes case-sensitive username)

TC-LOG-08	Case Sensitivity Test: Login with the correct username but a wrong password case.	User: testuser, Pass: password123!	Error message: "Invalid username or password." (Assumes case-sensitive password)
TC-LOG-09	Security Test: Attempt SQL Injection in the username field.	User: ' OR '1'='1, Pass: anyPassword	The system should sanitize input and return a generic "Invalid username or password" error.
TC-LOG-10	Account State Test: Login with correct credentials for a locked/inactive account.	User: lockeduser, Pass: Password123!	Error message: "This account has been locked. Please contact support."
TC-LOG-11	Functional Test: Verify the "Forgot Password?" link works.	Click the "Forgot Password?" link.	User is redirected to the password reset page.

4.1.1 Test Cases for Login

This table models the core authentication logic.

Rule	**1**	**2**	**3**	**4**	**5**
Conditions					
C1: User Found?	Y	Y	Y	N	Y
C2: Account Active?	Y	Y	N	-	N
C3: Password Match?	Y	N	-	-	Y
Actions					
A1: Login Successfully	X				

A2: Error: Invalid Creds		X		X	
A3: Error: Account Locked			X		X
Pass/Fail	**Pass**				

4.1.2 Decision Table for Login

Note on Precedence: *The dash - indicates a "don't care" condition. For example, if the user is not found (C1=N), it doesn't matter if the account is active or if the password matches. The system should always return the "Invalid credentials" error for security purposes (to not reveal if a username exists).*

4.2 Registration Test Case

This feature allows a new user to create an account. Inputs include username, email, password, and confirm password.

Test Case ID	Test Description	Test Data	Expected Result
TC-REG-01	Positive Path: Register with all new, valid, and matching data.	User: newuser, Email: new@example.com, Pass: Pass123!, Confirm: Pass123!	Account is created. User is redirected to a success page or logged in automatically.
TC-REG-02	Negative Path: Register with a username that already exists.	User: testuser, Email: new2@example.com, Pass: Pass123!	Error message: "Username is already taken."
TC-REG-03	Negative Path: Register with an email that already exists.	User: newuser2, Email: testuser@example.com, Pass: Pass123!	Error message: "An account with this email already exists."
TC-REG-04	Negative Path: Password and Confirm Password fields do not match.	User: newuser3, Email: new3@example.com, Pass: Pass123!, Confirm: Different!	Error message: "Passwords do not match."

TC-REG-05	Boundary Test: Register with a password that is too short (e.g., < 8 chars).	User: newuser4, Email: new4@example.com, Pass: short	Error message: "Password must be at least 8 characters long."
TC-REG-06	Validation Test: Register with a password that doesn't meet complexity rules (e.g., no number).	User: newuser5, Email: new5@example.com, Pass: password	Error message: "Password must contain at least one number and one uppercase letter."
TC-REG-07	Negative Path: Register with an invalid email format.	User: newuser6, Email: not-an-email, Pass: Pass123!	Error message: "Please enter a valid email address."
TC-REG-08	Negative Path: Leave a required field (e.g., Username) blank.	User: (blank), Email: new7@example.com, Pass: Pass123!	Error message: "Username is a required field."
TC-REG-09	Negative Path: Do not check the "Terms & Conditions" box.	All fields are valid, but "Terms & Conditions" is unchecked.	Error message: "You must accept the Terms & Conditions to register."
TC-REG-10	Security Test: Attempt XSS in the username field.	User: <script>alert('xss')</script>, Email: new8@example.com, Pass: Pass123!	The system should sanitize the input. The script should not execute. The username might be stored as alert('xss') or rejected.

4.2.1 Test Cases for Registration

This table focuses on the key validation rules that interact. Format and complexity checks are handled separately.

Rule	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**

Conditions																
C1: Username Unique?	Y	N	Y	Y	Y	N	N	Y	Y	N	Y	N	Y	N	Y	N
C2: Email Unique?	Y	Y	N	Y	Y	N	Y	N	Y	Y	N	N	Y	Y	N	N
C3: Passwords Match?	Y	Y	Y	N	Y	Y	N	Y	N	Y	Y	Y	N	N	N	N
C4: Terms Accepted?	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	Y	Y	Y	N
Actions																
A1: Create Successfully	X															
A2: Error: Username Taken		X				X	X			X		X		X		X
A3: Error: Email Taken			X			X		X			X		X		X	
A4: Error: Passwords Mismatch				X			X		X			X	X	X		X
A5: Error: Terms Not Accepted					X			X		X	X				X	X
Pass/Fail	**Pass**															

4.2.2 Decision Table for Registration

Note on Multiple Errors: In a real-world application, if a user submits a form with multiple errors (e.g., a taken username *and* a non-matching password), the system should ideally display all relevant error messages simultaneously. This decision table reflects that by allowing multiple 'X's in the action columns for a single rule (e.g., Rule 6, 7, 10, etc.).

4.3 Unsafe Conditions Submission

An "Unsafe Condition" refers to a state in the environment that has the potential to cause harm (e.g., a spill on the floor, a blocked fire exit, a frayed electrical cord).'

Test Case ID	Test Description	Test Data	Expected Result
TC-COND-01	Positive Path: Submit a complete and valid report.	Location: "Warehouse A", Description: "Pallet of boxes blocking fire extinguisher.", Severity: "Medium", Attachment: (photo.jpg)	The report is submitted successfully. A confirmation message is displayed. A unique report ID is generated.
TC-COND-02	Negative Path: Submit with a required field (Description) left empty.	Location: "Warehouse A", Description: (blank)	Error message: "Description is a required field." Submission is prevented.
TC-COND-03	Negative Path: Submit without selecting a location.	Location: (none selected), Description: "Leaking pipe in the ceiling."	Error message: "Location is a required field." Submission is prevented.
TC-COND-04	Boundary Test: Submit with an extremely long description (e.g., 5000 characters).	Description: (A very long string of text)	The system should either accept it or provide a clear error about the character limit.
TC-COND-05	File Upload Test: Submit with an invalid file type (e.g., .exe).	Attachment: malware.exe	Error message: "Invalid file type. Only JPG, PNG, and PDF are allowed."
TC-COND-06	File Upload Test: Submit with a file that exceeds the size limit (e.g., 15MB).	Attachment: large_photo.png (15MB)	Error message: "File size cannot exceed 10MB."

TC-COND-07	Functional Test: Click the "Cancel" or "Save as Draft" button.	Fill in some data, then click "Cancel".	The form is closed, and no data is submitted. The user is returned to the previous screen.
TC-COND-08	Permissions Test: Attempt to submit a report without being logged in.	User is not logged in.	User is redirected to the login page with a message: "Please log in to submit a report."

4.3.1 Test Cases for Unsafe Conditions Submission

This table models the core validation logic before a report can be submitted.

Rule	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Conditions								
C1: User Logged In?	Y	Y	Y	Y	N	N	N	N
C2: Location Selected?	Y	Y	N	N	-	-	Y	N
C3: Description Provided?	Y	N	Y	N	-	Y	-	N
Actions								
A1: Submit Successfully	X							
A2: Error: Login Required					X	X	X	X
A3: Error: Location Required			X	X				
A4: Error: Description Required		X		X		X		

Pass/Fail	**Pass**

4.3.2 Decision Table for Unsafe Conditions Submission

Note on Precedence: The system should first check if the user is logged in (C1). If not, it doesn't matter what other data is entered; the action is always "Error: Login Required." The dash - indicates a "don't care" condition.

4.4 Unsafe Actions Submission

An "Unsafe Action" refers to an observed behaviour of a person that could lead to an incident (e.g., an operator not wearing safety glasses, improper lifting, or bypassing a safety guard). This often includes an option to report anonymously.

Test Case ID	Test Description	Test Data	Expected Result
TC-ACT-01	Positive Path: Submit a complete, non-anonymous report.	Location: "Warehouse 3", Person Involved: "John Doe", Description: "Removed safety guard to clear jam.", Anonymous: (unchecked)	The report is submitted successfully, linking the action to "John Doe." Confirmation displayed.
TC-ACT-02	Positive Path: Submit a complete, anonymous report.	Location: " Warehouse 3", Person Involved: (blank), Description: "Removed safety guard to clear jam.", Anonymous: (checked)	The report is submitted successfully. The "Person Involved" field is stored as "Anonymous."
TC-ACT-03	Negative Path: Check "Anonymous" but still enter a name in "Person Involved."	Person Involved: "Jane Smith", Anonymous: (checked)	Error message: "Cannot specify a person if reporting anonymously. Please clear the name or uncheck anonymous."

TC-ACT-04	Negative Path: Do not check "Anonymous" and leave "Person Involved" blank.	Person Involved: (blank), Anonymous: (unchecked)	Report is submitted successfully. The system should treat this as an observation where the reporter did not know or wish to identify the person.
TC-ACT-05	Negative Path: Submit with a required field (Description) left empty.	Description: (blank)	Error message: "Description is a required field." Submission is prevented.
TC-ACT-06	Functional Test: Check the "Anonymous" box and verify the "Person Involved" field becomes disabled or cleared.	Check the "Anonymous" checkbox.	The "Person Involved" text field is either greyed out (disabled) or its contents are automatically cleared.
TC-ACT-07	Functional Test: Uncheck the "Anonymous" box and verify the "Person Involved" field becomes enabled.	Uncheck the "Anonymous" checkbox.	The "Person Involved" text field becomes active and can be typed in.

4.4.1 Test Cases for Unsafe Actions Submission

This table focuses on the specific business logic around anonymous reporting, assuming the user is logged in and a location is selected.

Rule	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Conditions								
C1: Description Provided?	Y	Y	Y	Y	N	N	N	N
C2: Anonymous Checked?	N	N	Y	Y	-	-	Y	N
C3: Person Involved Filled?	Y	N	N	Y	-	Y	N	Y
Actions								
A1: Submit (with name)	X	X						
A2: Submit (anonymously)			X					
A3: Error: Description Required					X	X	X	X
A4: Error: Anonymous Logic				X				
Pass/fail	**Pass**							

4.4.2 Decision Table for Unsafe Actions Submission

4.5 Manage Companies Module

This module allows administrators to perform CRUD (Create, Read, Update, Delete) operations on company records. The company record includes: Company ID, Company Name, Tax ID, Status (Active/Inactive), and Contact Information.

Test Case ID	Feature	Test Description	Test Data	Expected Result
TC-COMP-01	**Create**	**Positive Path:** Create a new company with all valid, unique data.	Name: "Acme Corp", Tax ID: "12-3456789", Status: Active	The company was created successfully and appears in the company list.
TC-COMP-02	**Create**	**Negative Path:** Attempt to create a company with a name that already exists.	Name: "Acme Corp" (already exists)	Error message: "Company with this name already exists." No new company created.
TC-COMP-03	**Create**	**Negative Path:** Attempt to create a company with a required field (e.g., Name) left empty.	Name: (blank), Tax ID: "98-7654321"	Error message: "Company Name is a required field."
TC-COMP-04	**Create**	**Boundary Path:** Attempt to create a company with an invalid Tax ID format.	Name: "Beta LLC", Tax ID: "12345"	Error message: "Invalid Tax ID format."
TC-COMP-05	**Update**	**Positive Path:** Update an existing company's information (e.g., change Status from Active to Inactive).	Change the status of "Acme Corp" to "Inactive"	The company record is updated. Change is reflected in the list and detail view.
TC-COMP-06	**Update**	**Negative Path:** Update a company's name to one that already exists for another company.	Change "Beta LLC" name to "Acme Corp"	Error message: "Company with this name already exists." Update is rejected.

TC-COMP-07	**Delete**	**Positive Path:** Delete a company that has no associated locations or records.	Delete "Gamma Inc."	Company is deleted and removed from the list.
TC-COMP-08	**Delete**	**Negative Path:** Attempt to delete a company that has associated locations.	Delete "Acme Corp" (which has locations)	Error message: "Cannot delete a company with existing locations."
TC-COMP-09	**Read/Search**	Search for a company by its exact name.	Search: "Acme Corp"	"Acme Corp" is displayed in the search results.
TC-COMP-10	**Read/Search**	Search for a company using a partial name (wildcard search).	Search: "Corp"	"Acme Corp" and other matching companies are displayed.

4.5.1 Test Cases for Manage Companies

This table focuses on the validation rules when creating a new company.

Rule	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Conditions								
C1: Name Unique?	Y	N	Y	Y	N	N	Y	N
C2: Tax ID Valid?	Y	Y	N	Y	N	Y	N	N
C3: Required Fields Filled?	Y	Y	Y	N	Y	N	N	N
Actions								
A1: Create Successfully	X							
A2: Error: Name Exists		X			X	X		X

A3: Error: Tax ID			X				X	
A4: Error: Missing Fields				X				
Pass/Fail	**Pass**							

4.5.2 Decision Table for "Create Company"

Note on Precedence: In cases with multiple errors (Rules 5, 6, 7, 8), the system's error handling order (precedence) will determine which message is shown. The table assumes "Name Exists" is checked first, followed by "Tax ID," then "Required Fields."

4.6 Manage Locations Module

This module manages physical locations, which are typically linked to a parent company. The location record includes: Location ID, Location Name, Address (City, State, Zip), and a link to the CompanyID.

Test Case ID	Feature	Test Description	Test Data	Expected Result
TC-LOC-01	**Create**	**Positive Path:** Create a new location with valid data linked to a valid company.	Name: "HQ", Address: "123 Main St", Company: "Acme Corp"	Location is created successfully and appears under "Acme Corp".
TC-LOC-02	**Create**	**Negative Path:** Attempt to create a location with a name that already exists *for the same company*.	Name: "HQ" (already exists for Acme Corp)	Error: "Location with this name already exists for this company."
TC-LOC-03	**Create**	**Negative Path:** Attempt to create a location with a name that exists for a *different* company.	Name: "HQ" (exists for Acme Corp), Company: "Beta LLC"	Location is created successfully (uniqueness is scoped to the parent company).

TC-LOC-04	**Create**	**Negative Path:** Attempt to create a location without selecting a parent company.	Name: "Branch Office", Company: (none selected)	Error: "Company is a required field."
TC-LOC-05	**Delete**	**Positive Path:** Delete a location that has no associated incidents or records.	Delete "Remote Office"	Location is deleted successfully.
TC-LOC-06	**Delete**	**Negative Path:** Attempt to delete a location that has associated incidents/records.	Delete "HQ" (which has incidents)	Error: "Cannot delete a location with associated records."
TC-LOC-07	**Update**	**Positive Path:** Update the address of an existing location.	Change "HQ" address to "456 Oak Ave"	Location is updated successfully.
TC-LOC-08	**View**	View the list of all locations for a specific company.	Navigate to "Acme Corp" -> Locations tab	Only locations associated with "Acme Corp" are displayed.

4.6.1 Test Cases for Manage Locations

Rule	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Conditions								
C1: Name Unique?	Y	N	Y	Y	N	N	Y	N
C2: Company Valid?	Y	Y	N	Y	N	Y	N	N
C3: Address Valid?	Y	Y	Y	N	Y	N	N	N
Actions								

A1: Create Successfully	X							
A2: Error: Name Exists		X			X	X		X
A3: Error: Company			X				X	
A4: Error: Address				X				
Pass/Fail	**Pass**							

4.6.2 Decision Table for "Create Location"

4.7 Manage Offense Codes Module

This module manages a master list of offense codes (e.g., theft, vandalism, assault). An offense code record includes: Offense Code (e.g., "THEFT-001"), Description, and Severity Level (e.g., Low, Medium, High).

Test Case ID	Feature	Test Description	Test Data	Expected Result
TC-OFF-01	Create	Positive Path: Create a new offense code with a unique code, description, and valid severity.	Code: "VANDAL-01", Desc: "Vandalism", Severity: "Medium"	Offense code is created successfully and appears in the list.
TC-OFF-02	Create	Negative Path: Attempt to create an offense code with a code that already exists.	Code: "THEFT-001" (already exists)	Error: "Offense Code already exists."
TC-OFF-03	Create	Negative Path: Attempt to create an offense code with an invalid severity level.	Code: "FRAUD-01", Severity: "Critical" (not an option)	Error: "Invalid Severity Level."

TC-OFF-04	Create	Negative Path: Attempt to create an offense code with a blank description.	Code: "ARSON-01", Desc: (blank)	Error: "Description is a required field."
TC-OFF-05	Delete	Positive Path: Delete an offense code that has never been used in an incident.	Delete code "TRESPASS-01"	Offense code is deleted successfully.
TC-OFF-06	Delete	Negative Path: Attempt to delete an offense code that has been used in one or more incident records.	Delete code "THEFT-001"	Error: "Cannot delete an offense code that is in use."
TC-OFF-07	Update	Positive Path: Update the description of an existing offense code.	Change "THEFT-001" description to "Petty Theft"	Offense code is updated successfully.
TC-OFF-08	Sort/Filter	Filter the offense code list by severity level "High".	Apply Filter: Severity = "High"	Only offense codes marked as "High" are displayed.

4.7.1 Test Cases for Manage Offense Codes

Rule	1	2	3	4	5	6	7	8
Conditions								
C1: Code Unique?	Y	N	Y	Y	N	N	Y	N
C2: Description Provided?	Y	Y	N	Y	N	Y	N	N
C3: Severity Valid?	Y	Y	Y	N	Y	N	N	N
Actions								
A1: Create Successfully	X							
A2: Error: Code Exists		X			X	X		X
A3: Error: Description			X				X	
A4: Error: Severity				X				
Pass/Fail	**Pass**							

4.7.2 Decision Table for "Create Offense Code"

5. Test Results and Analysis

This section presents a comprehensive summary of the results obtained from the primary testing phases: Black-Box Testing and Compatibility Testing.

5.1. Black-Box testing

A comprehensive Black-Box Testing phase was executed, covering several test cases designed to validate key system functionalities. The testing yielded a 100% success rate, with all actual outcomes precisely matching the expected results. This outcome provides strong validation that the developed system's functions are fully compliant with the specifications detailed in both the Software Requirement Specification (SRS) and the Software Design Document (SDD), confirming its alignment with stipulated user requirements.